

National University of Computer & Emerging Sciences

Quiz #4(a)

Name & Roll Number: _____

Date: _____

Section: _____

Maximum Marks: 10

1. A company has **three machines**, A, B, and C.

- Machine A produces **50%** of the items, Machine B produces **30%**, and Machine C produces **20%**.
- The defect rates are: Machine A: **2%**, Machine B: **4%**, Machine C: **5%**.

If an item is randomly selected and found to be defective, what is the probability that it was produced by Machine A? (Use Bayes' Theorem.)

2. There are three coins, identical in appearance, one of which is **ideal** and the other two are **biased** with probabilities $\frac{1}{3}$ and $\frac{2}{3}$ respectively for a head. One coin is taken at random and tossed twice. If a head appears both times, what is the probability that the ideal coin was chosen?

National University of Computer & Emerging Sciences

Quiz #4(b)

Name & Roll Number: _____

Section: _____

Date: _____

Maximum Marks: 10

1. A factory has 3 machines, Machine A produces 50% of the items, with a 1% defect rate. Machine B produces 30% of the items, with a 2% defect rate. Machine C produces 20% of the items, with a 5% defect rate. If an item is defective, what is the probability it was produced by Machine A?
2. A game provides two dice: one fair and one biased. The fair die has equal probability for all faces. The biased die rolls a 6 with probability $\frac{1}{2}$ and other values equally likely. A die is randomly chosen and rolled once, landing on 6. What is the probability that the fair die was chosen?

National University of Computer & Emerging Sciences

Quiz #4(c)

Name & Roll Number: _____
Date: _____

Section: _____
Maximum Marks: 10

1. A factory has **three production lines**, X, Y, and Z. X produces **40%** of the items, Y produces **35%**, and Z produces **25%**. Defect rates: X: **1.5%**, Y: **3%**, Z: **4%**. If an item is selected at random and found to be defective, what is the probability it was produced by Line Y?
2. A box contains three dice, one is fair (showing each number with probability $1/6$), while the other two are biased such that they land on a 6 with probabilities $1/3$ and $2/3$, respectively. One die is chosen at random and rolled twice .If a 6 appears both times, what is the probability that the fair die was chosen?

National University of Computer & Emerging Sciences

Quiz #4(d)

Name & Roll Number: _____
Date: _____

Section: _____
Maximum Marks: 10

1. A student takes a 4-option multiple-choice test:

- If they know the answer (80% chance), they get it right.
- If they guess (20% chance), they have a $\frac{1}{4}$ probability of being correct.

If the student answers correctly, what is the probability they knew the answer?

2. There are two coins, identical in appearance, one of which is **ideal** and the other is **biased** with probability $\frac{1}{3}$ for a head. One coin is taken at random and tossed twice. If a head appears both times, what is the probability that the ideal coin was chosen?